

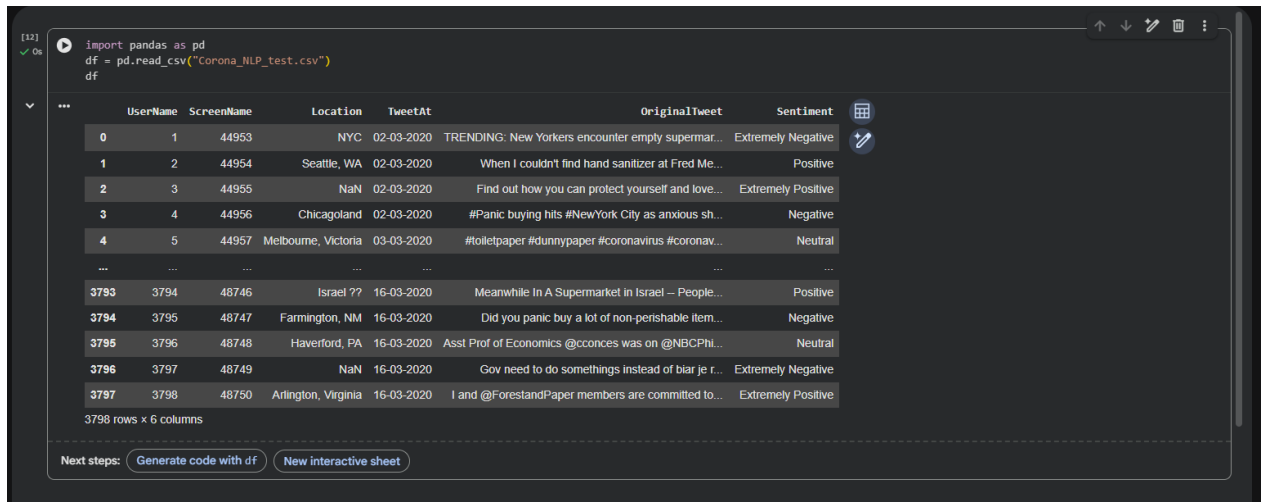
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PRN – 202501080011

Dataset – Kaggle text classification dataset

20 Problem Statement

1. Load dataset and show data



The screenshot shows a Jupyter Notebook cell with the following code:

```
[12] import pandas as pd
df = pd.read_csv("Corona_NLP_test.csv")
df
```

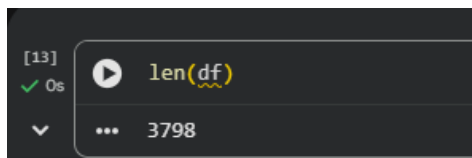
The output is a preview of the DataFrame with columns: Username, ScreenName, Location, TweetAt, OriginalTweet, and Sentiment. The preview shows rows 0 to 4 and rows 3793 to 3797. The Sentiment column contains values: Extremely Negative, Positive, Extremely Positive, Negative, Neutral, Positive, Negative, Neutral, Extremely Negative, and Extremely Positive.

	Username	ScreenName	Location	TweetAt	OriginalTweet	Sentiment
0	1	44953	NYC	02-03-2020	TRENDING: New Yorkers encounter empty supermar...	Extremely Negative
1	2	44954	Seattle, WA	02-03-2020	When I couldn't find hand sanitizer at Fred Me...	Positive
2	3	44955	NaN	02-03-2020	Find out how you can protect yourself and love...	Extremely Positive
3	4	44956	Chicagoland	02-03-2020	#Panic buying hits #NewYork City as anxious sh...	Negative
4	5	44957	Melbourne, Victoria	03-03-2020	#toiletpaper #dunnypaper #coronavirus #coronav...	Neutral
...	...	...	...	...	...	...
3793	3794	48746	Israel ??	16-03-2020	Meanwhile In A Supermarket in Israel – People...	Positive
3794	3795	48747	Farmington, NM	16-03-2020	Did you panic buy a lot of non-perishable item...	Negative
3795	3796	48748	Haverford, PA	16-03-2020	Asst Prof of Economics @cconces was on @NBCPhl...	Neutral
3796	3797	48749	NaN	16-03-2020	Gov need to do somethings instead of blar je r...	Extremely Negative
3797	3798	48750	Arlington, Virginia	16-03-2020	I and @ForestandPaper members are committed to...	Extremely Positive

3798 rows x 6 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

2. Total Tweets

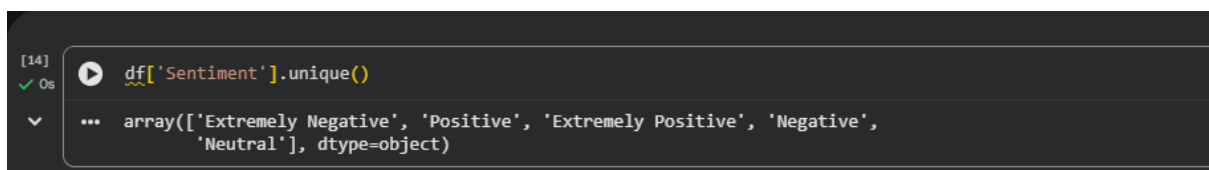


The screenshot shows a Jupyter Notebook cell with the following code:

```
[13] len(df)
```

The output is 3798.

3. Unique sentiments



The screenshot shows a Jupyter Notebook cell with the following code:

```
[14] df['Sentiment'].unique()
```

The output is an array of unique sentiments: ['Extremely Negative', 'Positive', 'Extremely Positive', 'Negative', 'Neutral'].

4. Count per sentiment

```
[15] df['Sentiment'].value_counts()
```

Sentiment	count
Negative	1041
Positive	947
Neutral	619
Extremely Positive	599
Extremely Negative	592

dtype: int64

5. Most frequent sentiment

```
[16] df['Sentiment'].value_counts().idxmax()
```

... 'Negative'

6. Least frequent sentiment

```
[17] df['Sentiment'].value_counts().idxmin()
```

... 'Extremely Negative'

7. Missing values

```
[18] df.isnull().sum()
```

UserName	0
ScreenName	0
Location	834
TweetAt	0
OriginalTweet	0
Sentiment	0

dtype: int64

8. Remove missing tweets

```
[30] df = df.dropna(subset=['OriginalTweet'])
     print(df.head())
```

	UserName	ScreenName	Location	TweetAt	OriginalTweet	Sentiment
0	1	44953	NYC	02-03-2020	trending: new yorkers encounter empty supermar...	Extremely Negative
1	2	44954	Seattle, WA	02-03-2020	when i couldn't find hand sanitizer at fred me...	Positive
2	3	44955	NaN	02-03-2020	find out how you can protect yourself and love...	Extremely Positive
3	4	44956	Chicagoland	02-03-2020	#panic buying hits #newyork city as anxious sh...	Negative
4	5	44957	Melbourne, Victoria	03-03-2020	#toiletpaper #dunnypaper #coronavirus #coronav...	Neutral

	length
0	228
1	193
2	73
3	318
4	252

9. Max length

```
[36] df['length'].max()
```

342

10. Min length

```
[37] df['length'].min()
```

19

11. Avg length per sentiment

```
[38] df.groupby('Sentiment')['length'].mean()
```

Sentiment	length
Extremely Negative	234.214527
Extremely Positive	240.268781
Negative	212.871278
Neutral	167.849758
Positive	213.923970

dtype: float64

12. Tweets with 'covid'

[39] `df[df['OriginalTweet'].str.contains('covid', na=False)]`

	UserName	ScreenName	Location	TweetAt	OriginalTweet	Sentiment	length
4	5	44957	Melbourne, Victoria	03-03-2020	#toiletpaper #dunnypaper #coronavirus #coronav...	Neutral	252
7	8	44960	Geneva, Switzerland	03-03-2020	@drtedros "we can't stop #covid19 without prot...	Neutral	214
9	10	44962	Dublin, Ireland	04-03-2020	anyone been in a supermarket over the last few...	Extremely Positive	238
10	11	44963	Boksburg, South Africa	04-03-2020	best quality couches at unbelievably low price...	Positive	233
11	12	44964	New Delhi	04-03-2020	beware of counterfeits trying to sell fake mas...	Extremely Negative	243
...	...	...	...	...	...	...	...
3785	3786	48738	People's Republic of China	16-03-2020	at this crucial time when non muslims are givi...	Extremely Positive	217
3786	3787	48739	Flemington, New Jersey	16-03-2020	we've noticed a shift in consumer #research in...	Extremely Positive	301
3794	3795	48747	Farmington, NM	16-03-2020	did you panic buy a lot of non-perishable item...	Negative	213
3796	3797	48749	NaN	16-03-2020	gov need to do somethings instead of biar je r...	Extremely Negative	174
3797	3798	48750	Arlington, Virginia	16-03-2020	I and @forestandpaper members are committed to...	Extremely Positive	254

2606 rows x 7 columns

13. Top words

[40] `df['OriginalTweet'].str.split().explode().value_counts().head(10)`

	count
the	4240
to	3723
and	2435
of	2060
in	1811
a	1801
#covid_19	1465
for	1341
is	1306
#coronavirus	1243

dtype: int64

14. Count total words in dataset

[42] `df['OriginalTweet'].str.split().apply(len).sum()`

... np.int64(124991)

15. First 5 tweets display

[43] `df['OriginalTweet'].head()`

	OriginalTweet
0	trending: new yorkers encounter empty supermar...
1	when i couldn't find hand sanitizer at fred me...
2	find out how you can protect yourself and love...
3	#panic buying hits #newyork city as anxious sh...
4	#toiletpaper #dunnypaper #coronavirus #coronav...

dtype: object

16. Count tweets starting with 'the'

```
[44] df['OriginalTweet'].str.startswith('the').sum()  
... np.int64(203)
```

17. Show column names

```
[45] df.columns  
... Index(['UserName', 'ScreenName', 'Location', 'TweetAt', 'OriginalTweet',  
          'Sentiment', 'length', 'Sentiment_num'],  
          dtype='object')
```

18. Total number of columns

```
[46] len(df.columns)  
... 8
```

19. Show last 5 tweets

```
[46] df['OriginalTweet'].tail()  
... 8
```

20. Count tweets with more than 50 characters

```
[47] (df['OriginalTweet'].str.len() > 50).sum()  
... np.int64(3782)
```

